

Practical Assignment

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Course: Enterprise Programming in Python Assessments
Book: Practical Assignment

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Practical Assignment

Faculty:	Information Technology
Module Code:	ITEPA3-33
Module Name:	Enterprise Programming in Python
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4.1. Discipline-specific Prohibited Activities

Please ensure that you consult the prohibited activity list that is specific to this module's discipline from the four discipline options below.



4.1.1. Software Engineering



4.1.2. Network Security Engineering



4.1.3. Data Science



4.1.4. Robotics

1. Assignment Overview

The rapid growth of digital learning, professional certification programmes and corporate skills development initiatives has transformed the education and training sector. Organisations now serve large numbers of learners across multiple locations while providing online enrolment, assessment services, learner support and operational reporting. As demand for these services continues to increase, organisations require enterprise applications that can process large volumes of transactions efficiently, support multiple users simultaneously and remain maintainable as business requirements evolve.

Many training providers continue to rely on ageing or monolithic systems that were originally designed for smaller user populations. As these systems expand, operational challenges begin to emerge. Registration processes become slower during peak enrolment periods, reporting functions consume increasing amounts of system resources, support requests accumulate faster than they can be processed and application maintenance becomes increasingly difficult. In many cases, poor scalability, ineffective concurrency management and tightly coupled system components result in reduced performance, limited flexibility and a poor user experience.

The organisation in this scenario provides professional certification programmes, short learning programmes and corporate training courses to thousands of learners across South Africa. The organisation currently uses several disconnected systems to manage learner registrations, course enrolments, assessments, support requests and reporting activities. As learner numbers continue to grow, management has identified concerns relating to application performance, duplicated information, inefficient workflows and the inability to scale individual components independently.

Management has therefore commissioned the development of a Python-based enterprise application prototype that will serve as a centralised platform for managing learner information, registrations, assessments, support requests and operational reporting. The solution must demonstrate how enterprise programming principles can be applied to improve scalability, maintainability, performance and reliability within a real-world environment.

The proposed system must support:

- learner registration and management
- course and assessment management
- support ticket creation and tracking
- concurrent registration processing
- enterprise reporting and monitoring
- performance optimisation
- automated testing
- application profiling
- error logging and diagnostics
- future migration to a microservices architecture

You are required to design and develop a Python-based enterprise application prototype that demonstrates the practical application of enterprise programming concepts. Your solution must reflect sound architectural decisions, appropriate use of design patterns, effective concurrency management, scalable data modelling techniques and software quality practices commonly found in modern enterprise systems.

Marks will be awarded for functionality, code quality, scalability considerations, testing, performance optimisation and overall solution design.

Deliverable 1

20 Marks

DELIVERABLE 1: 20 Marks

In this deliverable, you are required to design and implement the foundational components of the enterprise application. Your solution should demonstrate how enterprise applications use object-oriented principles and design patterns to improve maintainability, flexibility and scalability. The components developed in this deliverable will form the basis of the remaining sections of the project.

1.1 Domain Model Implementation (8 Marks)

Design and implement the following classes:

- Learner
- Course
- Registration
- Assessment
- SupportTicket

Your implementation should demonstrate sound object-oriented design principles and appropriate relationships between the classes. The design must include suitable attributes, methods, validation and class interactions that reflect the requirements of a large-scale training and certification environment.

The classes should support the management of learner information, course enrolments, assessment activities and support requests. You are encouraged to make reasonable assumptions where additional functionality may be required.

Provide screenshots showing that objects can be created successfully and that interactions between the classes function correctly.

Criteria	Marks
Appropriate class design and structure	2
Application of object-oriented principles	2
Relationships between classes	2
Validation and functionality	1
Evidence and screenshots	1
Total	8

1.2 Design Pattern Implementation (12 Marks)

Enterprise applications frequently use design patterns to improve maintainability, extensibility and code reuse. Implement the following design patterns within your application and demonstrate how they support enterprise application development.

Implement a Singleton Pattern to manage application configuration settings and ensure that only one configuration instance exists throughout the application. Implement a Factory Pattern that creates different support ticket types based on user requirements. Finally, implement a Strategy Pattern that supports different approaches for calculating assessment results.

For each design pattern, demonstrate how it has been incorporated into the application workflow and provide evidence that the implementation functions correctly. Briefly justify why each pattern is appropriate for the scenario and explain the benefits it provides to the application.

Provide screenshots showing the successful execution of each design pattern.

Criteria	Marks
Correct implementation of Singleton Pattern	3
Correct implementation of Factory Pattern	3
Correct implementation of Strategy Pattern	3
Justification and enterprise relevance	2
Evidence and screenshots	1
Total	12

Possible Output

```
=====
ENTERPRISE DOMAIN MODEL DEMONSTRATION
=====

DOMAIN MODEL OUTPUT
-----
Learner ID: L001, Name: Thabo Mokoena
Course: PY701 - Enterprise Python Development
Thabo Mokoena registered for Enterprise Python Development

SINGLETON PATTERN DEMONSTRATION
-----
Config 1 ID: 132090632838256
Config 2 ID: 132090632838256
Singleton Successful: Same Instance
```

```
FACTORY PATTERN DEMONSTRATION
-----
Created: AcademicTicket
Created: TechnicalTicket
Created: RegistrationTicket

STRATEGY PATTERN DEMONSTRATION
-----
Percentage Result: 82
Classification Result: Distinction
```

Deliverable 2

25 Marks

DELIVERABLE 2: 25 Marks

As the organisation continues to expand, the application must be capable of handling increasing numbers of learners and transactions without compromising performance or data integrity. In this deliverable, you are required to extend your application to support large-scale processing and concurrent operations that reflect real-world enterprise environments.

2.1 Registration Processing Engine (8 Marks)

Develop a registration processing engine that manages learner enrolments into courses. The registration engine should validate learner information, enforce business rules and maintain accurate registration records throughout the registration process. Your solution must prevent duplicate registrations, enforce course capacity limits and update registration statuses appropriately. The registration engine should also generate a summary showing the outcome of processed registrations, including successful and unsuccessful transactions.

To demonstrate scalability, your application must be capable of processing at least 10 simulated registration requests. Provide screenshots showing the execution and results produced by the registration engine.

Criteria	Marks
Registration processing logic	2
Validation and business rule enforcement	2
Duplicate registration prevention	1
Capacity management	1
Processing summary and output	1
Evidence and screenshots	1
Total	8

2.2 Scalable Data Modelling (5 Marks)

Evaluate the data structures used within your application and justify why they are suitable for processing at least 10 registration requests. Your discussion should compare at least two possible approaches and explain how your chosen implementation supports scalability, maintainability and performance.

Marking Guidelines

Criteria	Marks
Identification and discussion of data structures used within the application	1
Comparison of at least two possible data modelling approaches	1
Justification of the selected approach	1
Discussion of scalability, maintainability and performance considerations	1
Application of discussion to the implemented solution	1
Total	5

2.3 Concurrent Request Processing (12 Marks)

During peak registration periods, large numbers of learners may attempt to register simultaneously. Implement concurrent processing using an appropriate Python concurrency mechanism such as threading, asyncio or concurrent.futures. Your solution should demonstrate the processing of multiple registration requests at the same time while protecting shared resources and maintaining data integrity. Appropriate measures should be implemented to prevent inconsistent registrations and concurrency-related errors. Provide evidence of successful execution together with a brief explanation of how concurrency risks were identified and managed within the application.

Criteria	Marks
Correct implementation of concurrency mechanism	4
Processing of simultaneous requests	2
Identification of concurrency risks	2
Protection of shared resources and data integrity	2
Evidence and explanation	2
Total	12

Sample Output

REGISTRATION PROCESSING ENGINE

```
=====
[SUCCESS] Learner 1 registered
[SUCCESS] Learner 2 registered
[SUCCESS] Learner 3 registered
[SUCCESS] Learner 4 registered
[SUCCESS] Learner 5 registered
[SUCCESS] Learner 6 registered
[SUCCESS] Learner 7 registered
[SUCCESS] Learner 8 registered
[SUCCESS] Learner 9 registered
[SUCCESS] Learner 10 registered
```

REGISTRATION SUMMARY

```
-----
Total Registrations: 10
```

```
[SUCCESS] Concurrent Learner 2 registered
[SUCCESS] Concurrent Learner 11 registered
[SUCCESS] Concurrent Learner 6 registered
[SUCCESS] Concurrent Learner 10 registered
[FULL] Concurrent Learner 14 rejected. Course full.
[FULL] Concurrent Learner 4 rejected. Course full.
[FULL] Concurrent Learner 12 rejected. Course full.
[FULL] Concurrent Learner 7 rejected. Course full.
[FULL] Concurrent Learner 17 rejected. Course full.
[FULL] Concurrent Learner 16 rejected. Course full.
[FULL] Concurrent Learner 13 rejected. Course full.
[FULL] Concurrent Learner 5 rejected. Course full.
[FULL] Concurrent Learner 8 rejected. Course full.
[FULL] Concurrent Learner 9 rejected. Course full.
```

End of Question 3

Deliverable 3

20 Marks

Enterprise applications require effective monitoring and diagnostic capabilities to support operational visibility, troubleshooting and continuous improvement. In this deliverable, you are required to develop a monitoring subsystem called Bugzot that records application events, monitors performance and assists in identifying opportunities for optimisation.

3.1 Bugzot Monitoring Subsystem (8 Marks)

Develop a monitoring subsystem called Bugzot that records important events occurring within the application. The subsystem should assist administrators in identifying operational issues and monitoring overall system behaviour.

Your implementation should record validation failures, duplicate registration attempts, course capacity violations and other application errors generated during processing. Each recorded event should contain sufficient information to support troubleshooting and analysis.

Provide screenshots demonstrating the successful operation of the monitoring subsystem and examples of recorded events.

Criteria	Marks
Design and implementation of Bugzot subsystem	2
Logging of application events and errors	2
Quality and usefulness of captured information	2
Functionality and execution evidence	2
Total	8

3.2 Application Performance Monitoring (7 Marks)

Extend the Bugzot subsystem so that it records application performance information. The monitoring solution should provide visibility into the behaviour of the registration engine and other major application components. Your implementation should track transaction activity, processing performance and operational statistics that would assist management in evaluating system efficiency. Generate a performance report summarising the information collected by the monitoring subsystem.

Provide screenshots of the generated performance information and reports.

Criteria	Marks
Collection of performance metrics	2
Monitoring of transaction activity	2
Generation of performance reports	2
Evidence and screenshots	1
Total	7

3.3 Performance Improvement (5 Marks)

Enterprise applications often experience performance bottlenecks as usage increases. Analyse your application and identify one area where performance can be improved.

Explain the cause of the bottleneck, implement an improvement and compare the results before and after the optimisation. Support your discussion with evidence obtained from application output, monitoring information or performance measurements.

Criteria	Marks
Identification of performance issue	1
Analysis of root cause	1
Implementation of improvement	2
Evidence of improvement achieved	1
Total	5

Sample Output

```
=====
BUGZOT EVENT MONITORING LOG
=====
2026-06-19 14:40:01 | INFO | Registration Success | Anele Dlamini
    registered successfully.
2026-06-19 14:40:01 | INFO | Registration Success | Sipho Mokoena
    registered successfully.
2026-06-19 14:40:01 | ERROR | Validation Failure | Learner L003 has no name
    .
2026-06-19 14:40:01 | WARNING | Duplicate Registration | Learner L001 is
    already registered.
2026-06-19 14:40:01 | INFO | Registration Success | Lerato Nkosi registered
    successfully.
2026-06-19 14:40:01 | INFO | Registration Success | Thando Maseko
    registered successfully.
2026-06-19 14:40:01 | INFO | Registration Success | Naledi Khumalo
```

End of Deliverable 3

DELIVERABLE 4: 15 marks

4.1 User Interface Design and Evaluation

Management has indicated that a user-friendly front-end interface is required to support future expansion of the application. Develop a wireframe, storyboard, mock-up, or other suitable interface design showing how users would interact with the system.

Your design should demonstrate how learners and administrators would perform the following activities:

- Learner registration
- Course management
- Support ticket creation
- Report viewing

In your answer

1. Develop a suitable interface design that clearly represents the screens, navigation options, and user interactions required to support the activities listed above.
2. Briefly explain the design decisions used, including layout, navigation, usability, and validation considerations.
3. Evaluate your proposed design by discussing its strengths, identifying one limitation, and recommending an improvement.

Mark Allocation

Criteria	Marks
Completeness of interface design and inclusion of required functionality	5
Alignment of interface with application requirements	3
Explanation of design decisions (layout, navigation, validation, usability)	3
Evaluation of strengths and user experience considerations	2
Identification of limitation and recommendation for improvement	2
Total	15

End of Deliverable 4

DELIVERABLE 5: 20 Marks

Enterprise applications must be tested thoroughly, monitored continuously and designed with future growth in mind. In this deliverable, you are required to demonstrate software quality practices and evaluate how the application could evolve into a microservices-based solution.

5.1 Automated Testing (8 Marks)

Develop automated tests for key components of your application using an appropriate Python testing framework such as unittest or pytest.

Your tests should verify critical functionality within the application, including registration processing, business rule enforcement, assessment calculations and validation logic. The test suite should demonstrate that the application behaves as expected under normal operating conditions.

Provide screenshots showing the successful execution of the automated tests.

Criteria	Marks
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Selection and use of testing framework	2
Coverage of application functionality	3
Quality of test cases	2
Evidence and screenshots	1
Total	8

5.2 Application Profiling (4 Marks)

Use an appropriate Python profiling tool or technique to analyse the performance characteristics of your application.

Present the results obtained from the profiling activity and discuss what the findings reveal about the application's behaviour. Identify any areas that may require further optimisation.

Provide evidence of the profiling process and results.

Criteria	Marks
Use of profiling technique or tool	1
Analysis of profiling results	2
Identification of optimisation opportunities	1
Total	4

5.3 Microservices Readiness Assessment (8 Marks)

Analyse your existing application and identify how it could be decomposed into separate services.

Your proposal should:

- identify candidate services
- explain responsibilities
- discuss service communication
- explain testing and tracing requirements
- identify services capable of operating independently

Create a suitable architecture diagram to support your proposal.

Criteria	Marks
Identification of suitable candidate microservices	2
Explanation of service responsibilities and boundaries	2
Discussion of service communication mechanisms	1
Discussion of testing, tracing and monitoring considerations	1
Evaluation of independently executable services and associated benefits	1
Architecture diagram and overall presentation	1
Total	8

How AI use Affects Marking

Finding	Outcome
No AI used and accurately declared	Your lecturer will mark according to the marking guide.
Permitted AI use accurately declared	Your lecturer will mark according to the marking guide.
AI-generated in identifiable assessed component	Your lecturer will mark authentic work and will not award marks for the affected component that is identified to be AI-generated.
AI-generated in dependent criteria may be affected	Your lecturer will evaluate each criterion in isolation and will not award marks for affected criteria that are AI-generated.
Technically correct work that raises authenticity concerns*	Your lecturer will complete the review before finalising the mark, and will not award marks for affected criteria that are AI-generated.
Serious authenticity concerns exist*	Your lecturer may request a formal discussion with you, and depending on the outcome may escalate a case for formal academic-integrity hearing.

*See General Prohibited Activity list under Terms of AI use for this Assessment